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LSM

$$Y_{ij} = \alpha_i + \nu_j + \varepsilon_{ij}$$

$$\beta = \begin{pmatrix} \alpha_1 \\ \alpha_2 \\ \nu_1 \\ \nu_2 \end{pmatrix} \quad \mathbb{X} = \begin{bmatrix} 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{bmatrix}$$

To obtain, BLUE for $\underline{a}^T\underline{\beta}$ we observed that if $\underline{a}^T\underline{\beta}$ is estimable then $a^T \in \mathcal{R}(\mathbb{X})$ i.e.

$a^T\beta = u^T(\mathbb{X}\beta)$ for some $u \in \mathbb{R}^n$

i.e. $\underline{a}^T\underline{\beta} = \underline{u}^T\underline{\mu}$ where $\underline{\mu} = \mathbb{X}\underline{\beta}$

$$Y_{ijk} = \alpha_i + \nu_j + \varepsilon_{ijk}$$

$k = 1, \dots, m_{ij}$. $m_{ij} \geq 1 \ \forall \ i, j$

Then $\hat{\mu}_{LS} = \underset{\underline{\mu}}{\arg\min} \sum_i \sum_j \sum_k (Y_{ijk} - \mu_{ij})^2$

unconstrained least squares estimate of $\underline{\mu} = \begin{pmatrix} \mu_{11} \\ \mu_{12} \\ \mu_{21} \\ \mu_{22} \end{pmatrix}$

$$\hat{\mu}_{ij} = \frac{1}{m_{ij}} \sum_{k=1}^{m_{ij}} Y_{ijk}$$

If $\underline{a}^T\underline{\beta}$ is estimable then its unique blue is given by $\underline{u}^T\hat{\underline{\mu}}$ where u is s.t. $\underline{a} = \mathbb{X}^T\underline{u}$

Obtaining an estimate of μ is convenient since the corresponding least squares criterion is separable, i.e it can be expressed as a sum of quadratics in individual components of μ.

Consider the linear model

$$Y = \mathbb{X}\beta + \varepsilon$$

Normal Eqn. $(\mathbb{X}^T\mathbb{X})b = \mathbb{X}^T Y$

Solⁿ is most conveniently obtained if $\mathbb{X}^T\mathbb{X}$ is diagonal, i.e. the columns of $\mathbb{X}$ are mutually orthogonal.
This is an example of an orthogonal regression model.
In this case, the corresponding least squares estimate can be expressed in a separable form. i.e.,

$$\|Y - X\beta\|^2 = \|Y - \sum_{j=1}^{p} X_{\cdot j}\beta_j\|^2 \text{ where } \mathbb{X} = [X_{\cdot 1} \dots X_{\cdot p}]$$

$$= Y^TY - 2\sum_j \beta_j Y^T X_{\cdot j} + \sum_j \beta_j^2 X_{\cdot j}^T X_{\cdot j}$$

$\longrightarrow$ sum of quadratic functions in individual β_j's.

<u>Ex</u> Let there be r treatments $(r \geq 2)$ (eg drugs for treating hypertension)

$Y_{ij} \leftarrow$ outcome (eg systolic blood pressure level) of j^{th} subject (patient) receiving the i^{th} treatment.

$1 \leq j \leq m_i$

Is there any difference in the effect of the drugs?

Model: $Y_{ij} = \mu_i + \varepsilon_{ij}$ where ε_{ij}'s have independent and $\mathbb{E}\varepsilon_{ij} = 0$

For the time being, we also assume that $\mathrm{Var}\,\varepsilon_{ij} = \sigma^2$ for all i, j.

μ_i = mean of the i^{th} treatment group.

"Is there a difference in treatment effect?"

$$\mu_1 = \dots = \mu_r \;??$$

$\mu_i = \mu + \alpha_i$ where μ represents "overall mean"

and $\alpha_i = i^{th}$ treatment effect

This representation is meaningful only if we impose an identifiability constraint.

e.g. $\sum \alpha_i = 0$ or $\sum m_i \alpha_i = 0$

more generally $\sum_{i=1}^{r} w_i a_i = 0$ where $w_i \geq 0$

"Overall mean"

$$\text{(Sample) } \overline{Y}_{..} = \frac{\sum_i \sum_j Y_{ij}}{\underbrace{\sum_{i=1}^{r} m_i}_{n}} = \frac{\sum m_i \overline{Y}_{i.}}{n}$$

$$\mathbb{E}\overline{Y}_{..} = \frac{1}{n}\sum_i m_i \mathbb{E}\overline{Y}_{i.} = \frac{1}{n}\sum_i m_i \mu_i$$

$$= \frac{1}{n}\sum m_i(\mu + \alpha_i)$$

$$= \mu + \frac{\sum m_i \alpha_i}{n}$$

Matrix Representation

$$Y = \begin{pmatrix} Y_{11}\\ \vdots \\ Y_{1m_1}\\ Y_{21}\\ \vdots\\ Y_{2m_2}\\ \vdots\\ Y_{r1}\\ \vdots\\ Y_{rm_r}\end{pmatrix} = \underbrace{\begin{pmatrix}
1 & \left.\begin{matrix}1\\ \vdots\\ 1\end{matrix}\right\}m_1 & 0 & 0 & \cdots & 0\\
\vdots & 0 & \left.\begin{matrix}1\\ \vdots\\ 1\end{matrix}\right\}m_2 & 0 & \cdots & 0\\
\vdots & 0 & 0 & \left.\begin{matrix}1\\ \vdots\\ 1\end{matrix}\right\}m_3 & & 0\\
1 & 0 & 0 & 0 & \cdots & \left.\begin{matrix}1\\ \vdots\\ 1\end{matrix}\right\}m_r
\end{pmatrix}}_{\mathbb{X}} \underbrace{\begin{pmatrix}\mu\\ \alpha_1\\ \vdots\\ \alpha_r\end{pmatrix}}_{\beta} + \underline{\varepsilon}$$

Task: Eliminate α_r: i.e. express $\alpha_r = -\dfrac{\sum_{i=1}^{r-1} w_i \alpha_i}{w_r}$

$$Y = \mu X_{\cdot 1} + \alpha_1 X_{\cdot 2} + \dots + \alpha_{r-1} X_{\cdot (r-1)} + \alpha_r X_{\cdot r} + \underline{\varepsilon}$$

$$= \mu X_{\cdot 1} + \alpha_1 X_{\cdot 2} + \dots + \alpha_{r-1} X_{\cdot (r-1)} - \left(\sum_{i=1}^{r-1} \frac{w_i}{w_r}\alpha_i\right) X_{\cdot r} + \underline{\varepsilon}$$

$$= \mu X_{\cdot 1} + \sum_{i=1}^{r-1} \alpha_i \Big(\underbrace{X_{\cdot (i+1)} - \frac{w_i}{w_r} X_{\cdot (r+1)}}_{\tilde{X}_{\cdot (i+1)}}\Big) + \underline{\varepsilon}$$

$$\tilde{\mathbb{X}} = \left[X_{\cdot 1}\ \tilde{X}_{\cdot 2} \cdots \tilde{X}_{\cdot r}\right]$$

Two possible choices

$w_i = 1$ vs. $w_i = m_i$

Least squares estimate of $(\mu, \alpha_1, \dots, \alpha_r)$ is BLUE of $(\mu, \alpha_1, \dots, \alpha_r)$

$$\alpha_i = \mu_i - \mu \quad \mu = \frac{\sum w_i \mu_i}{\sum w_i}$$
$i = 1, 2, \dots, r$

when identifiability constraint is $\sum w_i \alpha_i = 0$

LSE of $\begin{pmatrix} \mu_1 \\ \vdots \\ \mu_r \end{pmatrix}$ is $\underset{u_1, \dots, u_r}{\operatorname{argmin}} \sum_{i=1}^{r} \sum_{j=1}^{m_i} (Y_{ij} - \mu_i)^2$

$\longrightarrow \hat{\mu}_i = \overline{Y}_{i\cdot}$
$i = 1, \dots, r$

BLUE for μ is $\dfrac{\sum w_i \hat{\mu}_i}{\sum w_i} = \hat{\hat{\mu}}$ say

BLUE for α_i is $\hat{\hat{\alpha}}_i = \hat{\mu}_i - \hat{\hat{\mu}}$, $i = 1, \dots, r$

LS estimation of $(\mu, \alpha_1, \dots, \alpha_r)$ when $w_i = m_i$ for $i = 1, \dots, r$
i.e. the constraint is $\sum m_i \alpha_i = 0$

Claim: In this case, the LSE of $(\mu, \alpha_1, \dots, \alpha_r)$ is

$$(\hat{\mu}, \hat{\alpha}_1, \dots, \hat{\alpha}_r) = (\overline{Y}_{\cdot\cdot}, \overline{Y}_{1\cdot} - \overline{Y}_{\cdot\cdot}, \dots, \overline{Y}_{r\cdot} - \overline{Y}_{\cdot\cdot})$$

LS Criterion

$$\sum_{i=1}^{r} \sum_{j=1}^{m_i} (Y_{ij} - \mu - \alpha_i)^2$$

$$= \sum_{i=1}^{r} \sum_{j=1}^{m_i} \left((Y_{ij} - \hat{\mu} - \hat{\alpha}_i) - (\mu - \hat{\mu}) - (\alpha_i - \hat{\alpha}_i) \right)^2$$

$$= \sum_{i=1}^{r}\sum_{j=1}^{m_i}(Y_{ij} - \underbrace{\hat{\hat{\mu}} - \hat{\hat{\alpha}}_i}_{\overline{Y}_{i\cdot}})^2 + \left(\sum_{i=1}^{r} m_i\right)(\mu - \hat{\hat{\mu}})^2$$

$$+ \sum_{i=1}^{r} m_i(\alpha_i - \hat{\hat{\alpha}}_i)^2 - 2(\mu - \hat{\hat{\mu}})\sum_i\sum_j \underbrace{(Y_{ij} - \overline{Y}_{i\cdot})}_{=0}$$

Notice:

$\sum m_i \alpha_i = 0$

$\sum m_i \hat{\hat{\alpha}}_i = 0$

$$- 2\sum_{i=1}^{m_i}(\alpha_i - \hat{\hat{\alpha}}_i)\sum_j \underbrace{(Y_{ij} - \overline{Y}_{i\cdot})}_{=0}$$

$$+ 2(\mu - \hat{\hat{\mu}})\sum_{i=1}^{m_i} \underbrace{m_i(\alpha_i - \hat{\hat{\alpha}}_i)}_{} = 0$$

Let $LHS = G(\mu, \alpha_1, \dots, \alpha_r)$

$RHS = \tilde{G}(\mu, \alpha_1, \dots, \alpha_r) = \sum\sum(Y_{ij} - \overline{Y}_i)^2$

$+ \sum m_i(\alpha_i - \hat{\hat{\alpha}}_i)^2$

$+ (\mu - \hat{\hat{\mu}})^2 (\sum m_i)$

① Under the constraint $\sum m_i \alpha_i = 0$ $\quad G(\ ) = \tilde{G}(\ )$

② $\underline{\text{Minimizer}}$ of $\tilde{G}$ is $(\hat{\hat{\mu}}, \hat{\hat{\alpha}}_1, \dots, \hat{\hat{\alpha}}_r)$ satisfies the constraint $\sum m_i \hat{\hat{\alpha}}_i = 0$
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